

# Dr. Chilperic Armel Foko Kuate



Computational scientist · Multiscale modeller

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Research website | [GitHub](#) | ORCID: 0000-0002-0140-7588

## Research profile

Applied mathematician and computational biologist working on mechanistic models across molecular, cellular and organismal scales. Research spans enzyme kinetics, nonlinear metabolic dynamics, stochastic immune-cell populations and plant-environment interactions. I connect model formulation to numerical simulation, parameter estimation, sensitivity, uncertainty and reproducible scientific software.

## Research experience

### Researcher · Computational Cell Biology

Oct 2026–present

Institute for Computational Cell Biology (CCB), Heinrich Heine University Düsseldorf

Research appointment resumed on 1 October 2026. Continuing work in computational biology and mechanistic modelling.

### Independent research & scientific software

Nov 2025–Sep 2026

Düsseldorf, Germany

Developed FokoLab and maintained research modelling projects; continued scientific writing and further training in AI engineering and research software.

### Postdoctoral researcher

Nov 2022–Oct 2025

CEPLAS / CCB, Heinrich Heine University Düsseldorf · Martin J. Lercher

- Developed a thermo-hydraulic-biochemical modelling framework for photosynthetic adaptation across the C3–C4 continuum, linking resource allocation, heat exchange, water transport and environmental forcing.
- Implemented simulation, parameter-exploration, optimisation and sensitivity workflows; worked with experimental collaborators and supervised student research.

### Marie Skłodowska-Curie doctoral researcher

Aug 2019–Aug 2022

PoLiMeR ITN / Institute of Quantitative and Theoretical Biology, HHU

- Built kinetic ODE models of hepatic fatty-acid synthesis and metabolism; investigated equilibria, stability and conditions for metabolic switching.
- Combined enzyme-kinetic literature, mathematical analysis and computational methods; contributed to peer-reviewed work on kinetic modelling and differentiable constraint-based models.

### Research fellow

Nov 2017–Mar 2019

African Institute for Mathematical Sciences · Cameroon and Rwanda

Developed deterministic, delay and stochastic T-cell population models and inference workflows for division-tracking data; taught mathematical and computational subjects.

## Selected research and software

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### FokoLab • simulation, inference and scientific visualisation

Browser-based scientific environment connecting custom models, dynamical systems, stochastic simulations, optimisation, sensitivity, fitting, statistics and mechanistic animation. Emphasis on recorded-state playback, explicit model scope and reproducible exports.

[Live platform](#) • [Source repository](#)

### Photosynthetic adaptation • mechanisms across scales

Plant physiology framework joining biochemical assimilation, heat balance, hydraulics, soil resources and climate forcing. Student and collaborative work includes Jérémie Muller-Prokob and Yvonne Danisch as main collaborators, Antonio Rigueiro and Jonas Brass, under Martin J. Lercher at CCB.

[Research code](#)

### Fatty-acid dynamics • kinetics and nonlinear behaviour

Doctoral modelling of de novo synthesis and metabolic pools, including the structure of steady states and possible switching regimes. Scientific collaboration with Oliver Ebenhöf, Adélaïde Raguin and Barbara M. Bakker.

[Synthesis model](#) • [Metabolic dynamics](#)

### T-cell proliferation • populations and lineages

Generation-structured models, Smith–Martin and Cyton formulations, and branching-process approaches to proliferation and loss. MSc research with Wilfred Ndifon and Gisèle Mophou; model-family provenance is kept distinct from later browser teaching implementations.

[Research code](#)

## Publications

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### Preprint • 2025

Hugo Dourado, **Chilperic Armel Foko Kuate**, Wolfram Liebermeister and Martin J. Lercher. *Growth Mechanics and the emergence of metabolic oscillations in growing cells*. bioRxiv. doi:10.1101/2025.06.24.661369. Preprint.

### Peer-reviewed review • 2023

**Chilperic Armel Foko Kuate**, Oliver Ebenhöf, Barbara M. Bakker and Adélaïde Raguin. *Kinetic data for modeling the dynamics of the enzymes involved in animal fatty acid synthesis*. Bioscience Reports 43(7), BSR20222496. doi:10.1042/BSR20222496.

### Peer-reviewed article • 2022

St. Elmo Wilken, Mathieu Besançon, Miroslav Kratochvíl, **Chilperic Armel Foko Kuate**, Christophe Trefois, Wei Gu and Oliver Ebenhöf. *Interrogating the effect of enzyme kinetics on metabolism using differentiable constraint-based models*. Metabolic Engineering 74, 72–82. doi:10.1016/j.ymben.2022.09.002.

## Education

### PhD • Quantitative and Theoretical Biology

2019–2023

Heinrich Heine University Düsseldorf, Germany. Completed 7 March 2023.

Dissertation: *Dynamic computational modeling of fatty acid de novo synthesis in the liver and qualitative analysis of fatty acid metabolism*. Supervision: Oliver Ebenhöf; co-supervision and collaboration: Adélaïde Raguin.

### MSc • Mathematics and Fundamental Applications

2017–2019

Modelling specialisation • University of Yaoundé I, Cameroon.

### MSc • Mathematical Sciences

2015–2016

AIMS Ghana / University of Cape Coast, Ghana.

### BSc • Mathematics

2009–2012

University of Douala, Cameroon.

### Baccalauréat

2009

Collège La Confiance, Bafoussam, Cameroon.

## Teaching, supervision and service

- Supervised two MSc and two BSc research projects in computational biology, metabolism and plant physiology.
- Delivered more than 200 hours of teaching and tutorials in mathematical modelling, differential equations, stochastic processes, algebra, scientific computing and Python across AIMS and HHU.
- Early Career Researchers' Representative, PoLiMeR ITN; earlier student representation at AIMS.

## Technical toolkit

**Programming:** Python, Julia, R, MATLAB; JavaScript for scientific interfaces.

**Methods:** ODE/DAE models, branching processes, nonlinear dynamics, fitting, uncertainty, local/global sensitivity and constrained optimisation.

**Practice:** NumPy, SciPy, SALib, CasADi/IPOPT, CMA-ES; Git, Linux, Jupyter, LaTeX, testing and reproducible environments.

## Selected training and professional development

|           |                                                                                                             |
|-----------|-------------------------------------------------------------------------------------------------------------|
| 2026      | AI-engineering and ML training in progress; scientific-software development; Bayer B4U mentoring programme. |
| 2023      | Optimal Control with CasADi, Leuven.                                                                        |
| 2021      | Advanced AI and Data Science Workshop, HeiCAD.                                                              |
| 2020–2022 | PoLiMeR doctoral training programme.                                                                        |
| 2018      | Epidemiological Modelling Clinic, AIMS South Africa / SACEMA.                                               |

## Selected recognition

Marie Skłodowska-Curie ITN fellowship, PoLiMeR; Post-AIMS Research Bursary; F.K.A. Allotey Meritorious Award, AIMS Ghana.

## Languages

French: native • English: proficient • German: B1/B1+, with B2 study in progress.